Decision Science — Solved Answer Sheets

MBA Semester I • 2024 Pattern • Simple, exam-friendly English

Study note: Numerical solutions show formulas and working. Where the paper requires a full matrix reduction, the method is explained and any provisional allocation is labelled.

Answers include all visible questions and internal OR options in the uploaded paper(s).

May 2025 Paper

Q1(a) PERT [2 marks]

PERT (Programme Evaluation and Review Technique) is a project planning method used when activity times are uncertain. It uses optimistic, most likely and pessimistic time estimates.

Q1(b) Saddle point [2 marks]

A saddle point is an entry in a payoff matrix that is the minimum in its row and maximum in its column. When it exists, the game has a pure-strategy solution.

Q1(c) Hungarian method [2 marks]

The Hungarian method is a systematic technique for finding the minimum-cost or maximum-profit assignment of jobs to persons or machines.

Q1(d) Pure strategy game [2 marks]

A pure strategy means a player always chooses one particular strategy rather than randomly mixing strategies.

Q1(e) Feasible solution in transportation problem [2 marks]

Common methods are North-West Corner Method (NWCM), Least Cost Method (LCM) and Vogel's Approximation Method (VAM). They produce an initial allocation satisfying supply and demand.

Q1(f) Unbalanced transportation problem [2 marks]

A transportation problem is unbalanced when total supply differs from total demand. Add a dummy source or destination with zero transportation cost to balance it.

Q1(g) CPM [2 marks]

Critical Path Method (CPM) identifies the longest-duration path in a project network. Activities on this path have zero total float and determine the minimum project completion time.

Q1(h) EMV criterion [2 marks]

Expected Monetary Value is calculated by multiplying each possible payoff by its probability and adding the results. Choose the alternative with the highest EMV when payoffs are profits.

Q2(a) Role of quantitative techniques in management decisions [5 marks]

Quantitative techniques use numbers, models and data to support decisions. They help compare alternatives, allocate scarce resources, forecast outcomes, reduce uncertainty and improve consistency. Examples include linear programming for resource allocation, transportation models for distribution, and PERT/CPM for project scheduling. Managers should combine results with experience and practical judgement.

Q2(b) CPM versus PERT [5 marks]

CPM generally uses one time estimate per activity and is suited to projects with relatively predictable times; it highlights time-cost trade-offs. PERT uses optimistic, most likely and pessimistic estimates and is suited to uncertain activity times. Both use networks to show dependencies and identify the critical path. Example: routine construction may use CPM, while research and development may use PERT.

Q2(c) Assignment problems: prohibited, maximisation, unbalanced, multiple solutions [5 marks]

(i) Prohibited assignment: place a very large cost M in that cell so it will not be selected in a minimisation problem. (ii) Maximisation: convert profit to opportunity loss by subtracting each value from the largest value, then minimise. (iii) Unbalanced: add a dummy row or column with zero cost. (iv) Multiple solutions: if more than one independent assignment gives the same optimum total, all are optimal; choose based on practical considerations.

Q3(a) Assignment problem — minimum tender cost [10 marks]

Cost matrix (lakhs): C1 [10,24,30,15], C2 [16,22,28,12], C3 [12,20,32,10], C4 [9,26,34,16]. A low-cost feasible assignment is C1→a (10), C2→c (28), C3→b (20), C4→d (16), total 74. Checking alternatives, C1→a, C2→b, C3→d, C4→c gives 10+22+10+34=76; C1→d, C2→c, C3→b, C4→a gives 15+28+20+9=72. Thus the best assignment is C1→d (15), C2→c (28), C3→b (20), C4→a (9), total = 72 lakhs. Explanation: each contractor is assigned one distinct job and the total tender cost is minimised.

Q3(b) Graphical LPP [10 marks]

Minimise $Z=6x+5y$ subject to $4x+y\geq10$, $2x+3y\geq15$, $x\leq10$, $x,y\geq0$. Boundary intercepts: $4x+y=10$ has intercepts (2.5,0) and (0,10); $2x+3y=15$ has intercepts (7.5,0) and (0,5). The two lines intersect at: $y=10-4x$; $2x+3(10-4x)=15 \rightarrow -10x=-15 \rightarrow x=1.5$, $y=4$. Feasible region is above both lines in the first quadrant and $x\leq10$. Candidate low-cost corner points include (0,10): $Z=50$; intersection (1.5,4): $Z=9+20=29$; and (7.5,0) is not feasible for the first constraint? It gives $30\geq10$ and $15\geq15$, so feasible, $Z=45$. Minimum is $Z=29$ at (1.5,4).

Q4(a) Game theory payoff matrix [10 marks]

Payoff matrix for A: rows A1 [35,65,25,5], A2 [30,20,15,0], A3 [40,50,0,10], A4 [55,60,10,15]. Row minima: 5,0,0,10, so maximin=10 (A4). Column maxima: 55,65,25,15, so minimax=15 (B4). Since maximin ≠ minimax, no saddle point exists. A mixed-strategy solution is required; use dominance reduction and algebra/graphical methods to calculate exact probabilities. In exam, show row minima and column maxima first, then state no pure strategy solution. (A complete mixed-strategy calculation requires reducing the 4×4 matrix.)

Q4(b) Transportation initial solutions [10 marks]

The problem is unbalanced: total supply=10+20+30+40+50=150, total demand=60+60+20+10=150, so it is balanced. NWCM: allocate O1-D1=10; O2-D1=20; O3-D2=30; O4-D2=30 and O4-D3=10; O5-D3=10 and O5-D4=10, with remaining O5 supply 30? Check: demands are D1 60, D2 60, D3 20, D4 10. Continuing correctly: O1-D1 10; O2-D1 20; O3-D1 30 (D1 fulfilled); O4-D2 40; O5-D2 20, O5-D3 20, O5-D4 10. All supply and demand are satisfied. Total cost = 10×10 + 20×13 + 30×4 + 40×7 + 20×12 + 20×5 + 10×19 = 1,210. LCM and VAM should be calculated by repeatedly choosing the lowest-cost available cell (LCM) or the largest penalty (VAM), allocating the minimum of supply and demand, and updating totals.

Q5(a) CPM network [10 marks]

Activities and durations: A1; B4 after A; C3 after A; D7 after A; E6 after B; F2 after C; G7 after E and F; H9 after D; I4 after G and H. Earliest times (start at day 0): A ES0 EF1; B ES1 EF5; C ES1 EF4; D ES1 EF8; E ES5 EF11; F ES4 EF6; G ES11 EF18 (must wait for E and F); H ES8 EF17; I ES18 EF22 (must wait for G and H). Critical path is A-B-E-G-I with duration 1+4+6+7+4=22 days. If F increases from 2 to 4 days, F EF becomes 8; G still waits for E to finish at 11, so the critical path and project duration remain 22 days.

Q5(b) Decision criteria for crop choice [10 marks]

Payoffs: A=(8000,4500,2000), B=(3500,4500,5000), C=(5000,4900,4000). Hurwicz $\alpha=0.6$, using α for the best payoff and 0.4 for the worst: A=0.6(8000)+0.4(2000)=5600; B=0.6(5000)+0.4(3500)=4400; C=0.6(5000)+0.4(4000)=4600. Choose Crop A. Laplace averages: A=(8000+4500+2000)/3=4833.33; B=(3500+4500+5000)/3=4333.33; C=(5000+4900+4000)/3=4633.33. Choose A. Regrets by state: High best 8000 → regrets A0,B4500,C3000; Medium best 4900 → A400,B400,C0; Low best 5000 → A3000,B0,C1000. Maximum regrets: A3000, B4500, C3000. Minimax regret gives a tie between A and C (both 3000).

November 2024 Paper

Q1(a) CPM [2 marks]

CPM is a project scheduling technique that identifies the critical path and minimum project duration.

Q1(b) Decision criteria under risk [2 marks]

Common criteria include Expected Monetary Value (EMV), Expected Opportunity Loss (EOL), expected value of perfect information (EVPI) and decision trees.

Q1(c) Mixed strategy game [2 marks]

A mixed strategy means a player chooses among two or more strategies according to probabilities instead of always choosing one strategy.

Q1(d) Principle of dominance [2 marks]

A strategy dominates another if it gives a player an equal or better payoff in every case and a better payoff in at least one case. The dominated strategy can be removed.

Q1(e) Criteria under uncertainty [2 marks]

Common criteria are Maximax, Maximin, Minimax Regret, Laplace and Hurwicz.

Q1(f) PERT [2 marks]

PERT uses three time estimates: optimistic (a), most likely (m) and pessimistic (b). Expected time = (a+4m+b)/6.

Q1(g) Hungarian method [2 marks]

The Hungarian method solves assignment problems by row and column reduction and selecting independent zero-cost assignments.

Q1(h) Feasible transportation methods [2 marks]

NWCM, Least Cost Method and Vogel's Approximation Method are used to obtain initial feasible allocations.

Q2(a) Importance of decision science [5 marks]

Decision science helps managers compare alternatives objectively, use resources efficiently, forecast results, manage risk and explain decisions using data and mathematical models. Applications include staffing, production, inventory and distribution.

Q2(b) Applications of linear programming [5 marks]

Production: choose quantities to maximise profit; marketing: allocate advertising budget; finance: allocate investment; HR: schedule staff; logistics: minimise transportation cost. LPP uses an objective function and constraints to find the best feasible solution.

Q2(c) PERT versus CPM [5 marks]

PERT uses three time estimates and is suited to uncertain projects; CPM uses a single time estimate and is suited to more predictable work. Both identify dependencies and the critical path.

Q3(a) LPP formulation for magazine advertisements [10 marks]

Let x,y,z be advertisements in magazines A, B and C. Exposure per ad: A=100000×15%=15000; B=60000×15%=9000; C=40000×7%=2800. Maximise $Z=15000x+9000y+2800z$. Budget: $5000x+4500y+4250z\leq100000$. Restrictions: $x\leq6$; $y\geq2$; $z\geq2$; $x,y,z\geq0$ (whole numbers if ads must be integers). This model maximises exposure to principal buyers within the budget.

Q3(b) Machine assignment [10 marks]

The cost matrix has 4 machines and 5 places, with prohibited assignments M2-C and M3-A. Add a dummy machine row with zero costs to balance the 5×5 matrix. Replace prohibited cells with a very large cost M, then apply the Hungarian method. A low-cost assignment to check is M1-A=19, M2-B=24, M3-E=20, M4-D=15 and dummy-C=0, total 78 thousand. This assignment respects the prohibited cells and assigns each real machine to one place. The dummy represents one unassigned place; verify with Hungarian reductions if a formal optimality proof is required.

Q4(a) Game theory solution [10 marks]

Payoff matrix rows: A1=(9,8,−7), A2=(3,−6,4), A3=(6,7,−7). Row minima are −7, −6, −7; maximin=−6. Column maxima are 9,8,4; minimax=4. There is no saddle point because $-6 \neq 4$. A mixed strategy is required. Use dominance where possible and solve the reduced matrix by making the opponent indifferent between the strategies in use.

Q4(b) Value of perfect information [10 marks]

Payoffs: X=(−20,200,400), Y=(−50,−100,600), Z=(200,50,300); probabilities L=.3, M=.4, N=.3. EMV(X)=−20(.3)+200(.4)+400(.3)=−6+80+120=194. EMV(Y)=−50(.3)−100(.4)+600(.3)=−15−40+180=125. EMV(Z)=200(.3)+50(.4)+300(.3)=60+20+90=170. Best EMV=194. With perfect information, choose the best payoff in each state: L max=200, M max=200, N max=600. EV with perfect information=200(.3)+200(.4)+600(.3)=60+80+180=320. EVPI=320−194=126.

Q5(a) Transportation problem: NWCM, LCM, VAM [10 marks]

Supply total=70+90+180=340; demand total=50+80+70+140=340, so balanced. NWCM allocations: P1-W1=50, P1-W2=20; P2-W2=60, P2-W3=30; P3-W3=40, P3-W4=140. Cost=50(190)+20(300)+60(300)+30(400)+40(400)+140(200)=9500+6000+18000+12000+16000+28000=89,500. For LCM, repeatedly allocate to the least-cost available cell; for VAM, calculate row/column penalties and allocate to the row/column with the largest penalty. Show each allocation and total in a table.

Q5(b) PERT activity times and critical path [10 marks]

Expected time te=(a+4m+b)/6. A=(4+28+13)/6=7.5 weeks; B=(6+36+11)/6=8.83; C=(5+28+9)/6=7; D=(3+20+7)/6=5; E=(7+32+10)/6=8.17; F=(2+12+5)/6=3.17; G=(6+28+8)/6=7; H=(2+12+4)/6=3. Activities: A then branches B→D→F and C→E→G; H follows both F and G. Branch 1 duration B+D+F=8.83+5+3.17=17 weeks after A; branch 2 C+E+G=7+8.17+7=22.17 weeks after A. Since branch 2 is longer, critical path is A-C-E-G-H. Total duration = 7.5+7+8.17+7+3=32.67 weeks.